Mathematics: analysis and approaches
Standard level
Paper 1 — markscheme
4.1–4.6 Statistics and probability

13 October 2026

W4ss Academy practice set 1 | **Topic** 4.1–4.6 Statistics and probability | **Ref.** W4SS–MAA–T4–P1–01

Student name

Paper 1

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- **M**: method marks. **A**: answer or accuracy marks, usually after the M mark they depend on. **R**: reasoning marks. **N** (former course): marks for a correct answer with no working.
- Marks in brackets, e.g. **(M1)**, can be implied by correct working that follows. **AG**: answer given in the question, no mark for it. **FT**: follow through from an earlier error.
- Total: **[25 marks]**.

W4ss Academy 2026

Section A

Question	Marks	Markscheme
1. (a)	**[1]**	$q = 5$ ***A1 N1***
1. (b)	**[2]**	valid approach ***(M1)*** *eg* $(18 + 10 + 5) - 30$, $28 - 25$, $18 + 10 - n = 25$ $n = 3$ ***A1 N2***
1. (c)	**[3]**	valid approach for finding m or p (may be seen in part (b)) ***(M1)*** *eg* $18 - 3$, $3 + p = 10$ $m = 15$, $p = 7$ ***A1A1 N3***
2. (a)	**[3]**	valid approach to find P(R) ***(M1)*** tree diagram (must include probabilty of picking box) with correct required probabilities OR $P(R \cap B_1) + P(R \cap B_2)$ OR $P(R \mid B_1)P(B_1) + P(R \mid B_2)P(B_2)$ $\frac{5}{7} \cdot \frac{1}{2} + \frac{4}{7} \cdot \frac{1}{2}$ ***(A1)*** $P(R) = \frac{9}{14}$ ***A1***
2. (b)	**[2]**	events A and R are not independent, since $\frac{9}{14} \cdot \frac{1}{2} \neq \frac{5}{14}$ OR $\frac{5}{7} \neq \frac{9}{14}$ OR $\frac{5}{9} \neq \frac{1}{2}$ OR an explanation e.g. different number of red balls in each box ***A2*** **Note:** Both conclusion and reasoning are required. Do not split the ***A2***.

Section B

Question	Marks	Markscheme
3. (a)	**[2]**	evidence of median position ***(M1)*** 40 students median = 14 (hours) ***A1***
3. (b)	**[3]**	recognizing there are 8 students in the top 10% ***(M1)*** 72 students spent less than k hours ***(A1)*** $k = 18$ (hours) ***A1***
3. (c)	**[4]**	15 hours is 60 students OR $p = 60 - 4$ ***(M1)*** $p = 56$ ***A1*** 21 hours is 76 students OR $q = 80 - 76$ OR $q = 80 - 4 - 56 - 16$ ***(A1)*** $q = 4$ ***A1***

3. (d)	**[3]**	20 of the 80 students OR $\frac{1}{4}$ spend more than 15 hours doing homework ***(A1)*** $\frac{20}{80} = \frac{x}{320}$ OR $\frac{1}{4} \times 320$ OR 4×20 ***(A1)*** 80 (students) ***A1***
3. (e)(i)	**[1]**	only year 12 students surveyed OR amount of homework might be different for different year levels ***R1***
3. (e)(ii)	**[1]**	stratified sampling OR survey students in all years ***R1***

Please **do not** write on this page.

Answers written on this page
will not be marked.